Mohamed Sameh Salah Eldine

Data Scientist | Machine Learning Engineer
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SUMMARY

Data Scientist and ML Engineer with hands-on experience in deep learning, computer vision, LLM agents and streaming data pipelines using Python, PyTorch and TensorFlow. Built a streaming manufacturing analytics platform (Kafka, Java, C++, SQL Server, dbt, R), an MCP-based agent system, a Transformer and Bi-LSTM trading model with walk-forward backtesting, and an ICAO-compliant passport OCR pipeline. Completed field training in data science and AI engineering at Comply Market. B.Sc. in Computing and Data Science (GPA 3.50 / 4.00).

TECHNICAL SKILLS

Languages: Python, R, SQL, TypeScript, JavaScript, C++, Java
Machine Learning: PyTorch, TensorFlow, scikit-learn, Transformers, LSTM, statistical analysis, predictive modeling, data cleaning, feature engineering, hyperparameter optimization (Optuna), model evaluation, walk-forward backtesting
Computer Vision and NLP: OpenCV, EasyOCR, Tesseract, spaCy, semantic segmentation (U-Net)
LLMs and Agents: LLM APIs (Claude, Gemini, OpenAI), Model Context Protocol (MCP), prompt engineering, agentic workflows, Amazon Bedrock, Strands Agents
Data and Visualization: Pandas, NumPy, ETL pipelines, PostgreSQL, MongoDB, SQL Server, SQLite, Power BI, Tableau, Streamlit, Plotly, Matplotlib, Seaborn
Backend and Cloud: FastAPI, Node.js, REST APIs, Docker, Kafka, GitHub Actions (CI/CD), AWS (S3, Bedrock), Google Cloud (Cloud Run), Git, Linux

EXPERIENCE

Field Training, Data Science and AI Engineering Aug 2026 – Oct 2026
Comply Market, Alexandria, Egypt

- Supported the team on data science and AI engineering tasks across ongoing projects.
- Contributed to building an AI agent integrated into the company website.

Data Science Training Sep 2025
MAIM Digital Solutions

- Completed hands-on training in machine learning workflows, data analytics and project implementation.

IoT Field Training Jul 2024 – Sep 2024
Smart Garage System

- Built a Smart Garage System on ESP32 and Arduino with a Flutter monitoring app, using MQTT and Firebase for real-time cloud synchronization.

PROJECTS

FactoryPulse: Streaming Manufacturing Analytics Platform Oct 2026
Kafka, Java, C++, TensorFlow, R, SQL Server, dbt, Docker | Code: github.com/medo-salah/factorypulse

- Built an end-to-end streaming platform on a synthetic predictive-maintenance dataset (AI4I 2020, 10,000 records, 20 simulated machines): a C++17 edge agent replays events into Apache Kafka, a Java 21 Kafka Streams service applies rule alerts and one-minute window aggregates, and an idempotent JDBC sink writes to SQL Server.
- Modeled the data as a star schema with dbt on SQL Server (staging, marts, reporting views, slowly changing dimension snapshot) with automated data tests and read-only views for BI tools.
- Compared logistic regression, random forest, gradient boosting and a TensorFlow network using leakage controls, validation-set thresholds, bootstrap confidence intervals and paired tests; deterministic rules outperformed the best ML model on this synthetic data (F1 0.90 vs 0.86).
- Added a TensorFlow LSTM autoencoder for anomaly detection and R and Quarto reports covering control charts, process capability, and McNemar and DeLong tests.
- Benchmarked the C++ rolling-statistics engine at about 560,000 events per second (5 channels, 50-sample window, Apple A18 Pro); containerized the stack with Docker Compose and GitHub Actions CI (sanitizer-checked C++ tests, Java, Python and R tests, secret scanning).

Threadback: Intent-Recovery AI Agent Oct 2026
Python, FastAPI, MCP, React, TypeScript | Code: github.com/medo-salah/Threadback_AWS | Demo: threadback-aws.vercel.app

- Built an open-source agent system that reconstructs unfinished user commitments from conversation context, exposing 9 tools through a Model Context Protocol (MCP) server over Streamable HTTP.
- Designed a deterministic verification layer so the LLM cannot mark a task complete without evidence; integrated an Amazon Bedrock and Strands Agents provider with SQLite persistence and a React interface showing live tool activity.
- Verified the system with 401 automated tests, linting and a clean production build.

AI Candlestick Trader 2026

PyTorch, Transformers, LSTM, Optuna, Streamlit | Code: github.com/medo-salah/ai-candlestick-trader

- Built a deep-learning pipeline for EGX and Tadawul stocks with 22 technical features and 18 candlestick-pattern detectors engineered from OHLC data.
- Trained Transformer and attention-based Bi-LSTM models with inverse-variance-weighted ensembling and Optuna search; used walk-forward backtesting to avoid data leakage (RMSE, Sharpe ratio, Calmar ratio, win rate).
- Delivered an interactive Streamlit and Plotly dashboard and a pytest suite.

Passport OCR and Document Verification 2026

FastAPI, EasyOCR, OpenCV, spaCy | Code: github.com/medo-salah/passport_ocr

- Built an OCR pipeline combining EasyOCR, Tesseract, OpenCV and spaCy to extract and validate passport and ID data.
- Implemented ICAO-compliant MRZ detection and parsing, with Error Level Analysis for authenticity checks and face-quality analysis.
- Exposed the pipeline through a FastAPI REST API with batch processing and an analytics dashboard.

AI Insight Visualizer 2026

React, TypeScript, Express, PostgreSQL, Redis, BullMQ | Code: github.com/medo-salah/AI-Insight-Visualizer

- Built a full-stack analytics platform that generates AI-driven insights, reports and visualization recommendations from uploaded datasets using OpenAI services.
- Designed a type-safe monorepo with OpenAPI specifications and Zod validation, with background jobs on BullMQ and Redis.

Hasibha: Personal Finance Analytics Platform (Graduation Project) 2026

Node.js, Express.js, MongoDB | Alexandria University

- Built the data and analytics layer of an Arabic-first personal finance platform for Egyptian users, covering transactions, saving goals with progress tracking and multi-account management.
- Designed a MongoDB Aggregation Pipeline engine that produces the financial statistics and dashboard analytics.
- Enforced user-scoped queries across all modules so each user's financial data stays isolated, and secured access with JWT authentication, input validation and rate limiting.

Hexaverse Authentication Backend 2025

Node.js, Express.js, MongoDB | Code: github.com/medo-salah/hexaverse-auth-backend

- Built the secure access layer that data-driven applications sit on, supporting Email/Password and Google OAuth sign-in with JWT.
- Implemented password recovery, email verification and rate limiting behind scalable APIs for mobile and web clients.

Nagah: Multi-Tenant EdTech SaaS (in development) 2026

NestJS, React, TypeScript, PostgreSQL, Docker

- Designing a multi-tenant platform for Egyptian secondary-education teachers with role-based access control, payment review workflows and Docker-based deployment.
- Integrated Claude through a server-side adapter for AI essay grading against rubrics and lesson and quiz generation.

LoveDA Semi-Supervised Segmentation 2026

PyTorch, U-Net | Code: github.com/medo-salah/LoveDA-SemiSupervised-Segmentation

- Trained a semantic segmentation model using sparse point-level supervision and a custom Partial Cross-Entropy loss function.
- Optimized training for low-resource GPU environments.

EDUCATION

B.Sc. in Computing and Data Science Sep 2022 – Jun 2026

Faculty of Computers and Data Science, Alexandria University

GPA: 3.50 / 4.00. **Relevant coursework:** Machine Learning, Artificial Intelligence, Data Mining, Database Systems, Cloud Computing, Data Structures, Algorithms, Operating Systems, Computer Networks, Web Development.

CERTIFICATIONS AND COURSES

Certifications: AWS Certified AI Practitioner (Manara Learning Path); Cisco Certified Network Associate (CCNA); Supervised Machine Learning: Regression and Classification (DeepLearning.AI)

Courses: Junior Cybersecurity Analyst Career Path and Cyber Threat Management (Cisco Networking Academy); Understanding Data Science and Exploratory Data Analysis in Python (DataCamp); Software Engineering Level 1 (Orange Digital Center); CIB Summer Program (The Green Leap)

LEADERSHIP AND VOLUNTEERING

Chief of the Movement Member for 12 years

Saint Vincent Miami Scouts

- Lead a group of 40 members aged 17, planning and running camping trips and human-development sessions.

LANGUAGES

Arabic (Native) | English (B2) | French (B2)